Quantifying comprehensive marine vessel emissions using data fusion

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1 In-line manuscript statistics

- The transportation sector is a major source of global greenhouse gas (GHG) and local air pollution emissions, responsible for 21% of all CO_2 emissions globally in 2023
- For over 900,000 AIS-broadcasting vessels, we quantify point-estimate emissions for each vessel using precise latitude/longitude/timestamp information for over 97 billion individual AIS messages

2 Figures

2.1 Figure 1

2.1.1 Figure 1, panel A

2.1.2 Figure 1, panel B

2.1.3 Figure 1, combined

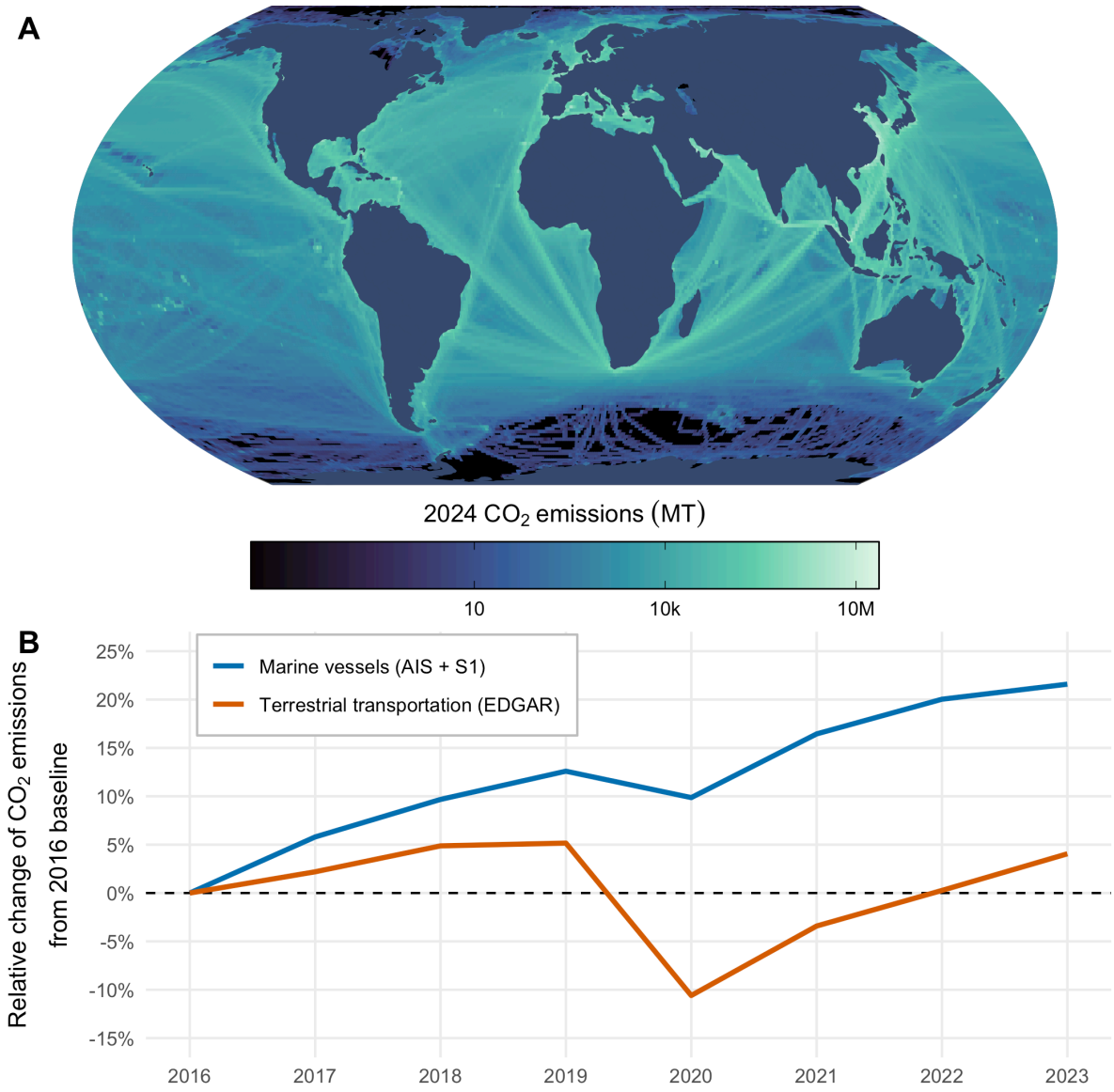

Figure 1

2.2 Figure 2

2.2.1 Figure 2, panel A

2.2.2 Figure 2, panel B

2.2.3 Figure 2 combined

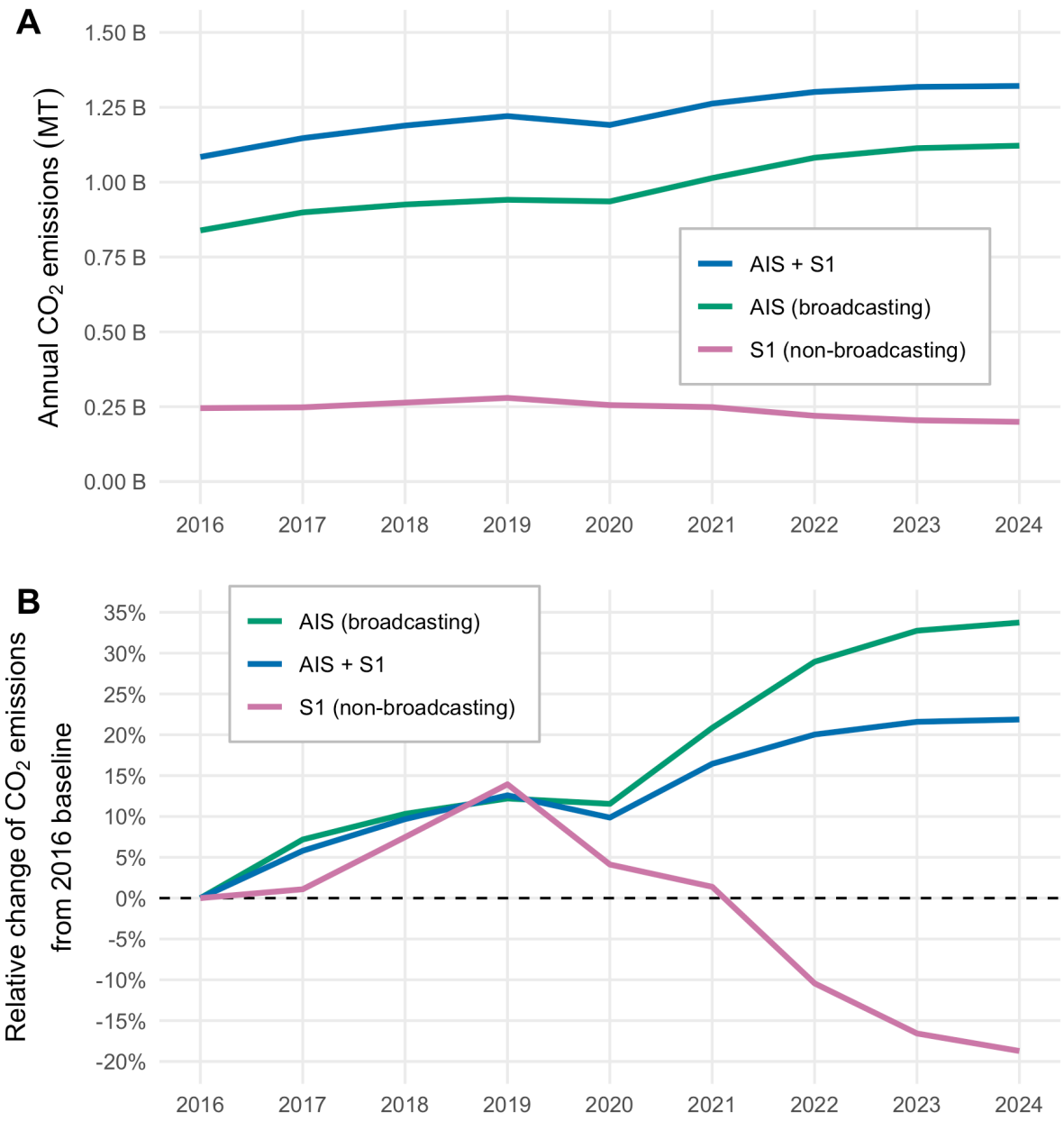

Figure 2: (A) Annual marine vessel CO2 emissions, with each line colored based on the data source. (B) Relative change in annual marine vessel CO2 emissions from a 2016 baseline, with each line colored based on the data source.

2.3 Figure 3

Table 1: Performance metrics for random forest model that predicts annual CO2 emissions, by vessel size class and type, using hours and average speed.

.metric	.estimate
rsq	0.995
mape	8.253
rsq_trad	0.995

2.4 Figure 3

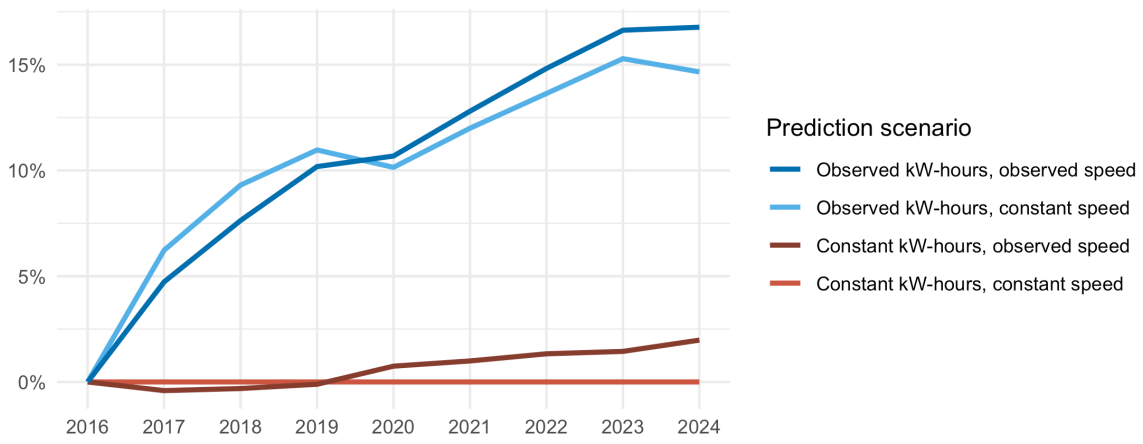

Figure 3: Estimated decomposition of change in total CO2 emissions from 2016 baseline (aggregating across both broadcasting and non-broadcasting vessels), using the random forest model outputs of four different scenarios that decompose the relative impact of changes in total kW-hours and changes in average speed/

3 Supplementary figures

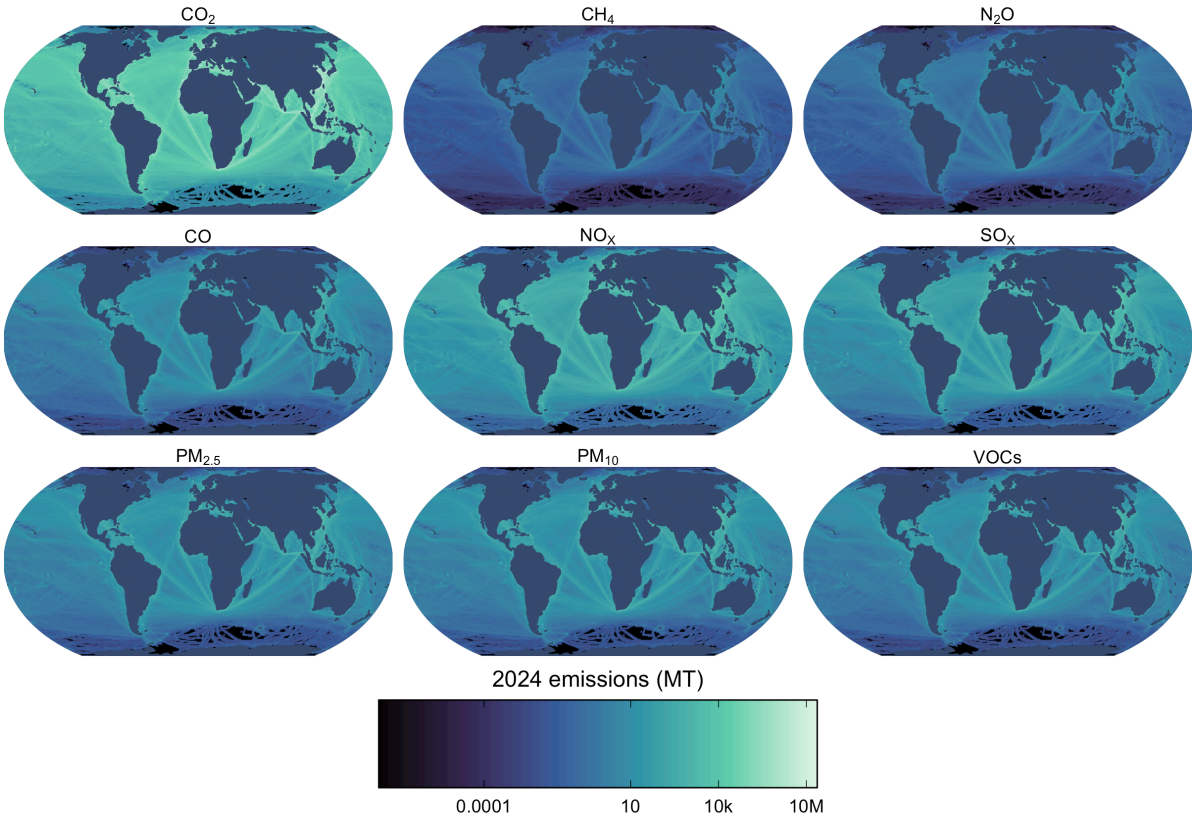

Figure 4: Spatial map of 2024 emissions for each pollutant, aggregated across the AIS-broadcasting fleet and non-broadcasting, and shown at a 1x1 degree spatial resolution

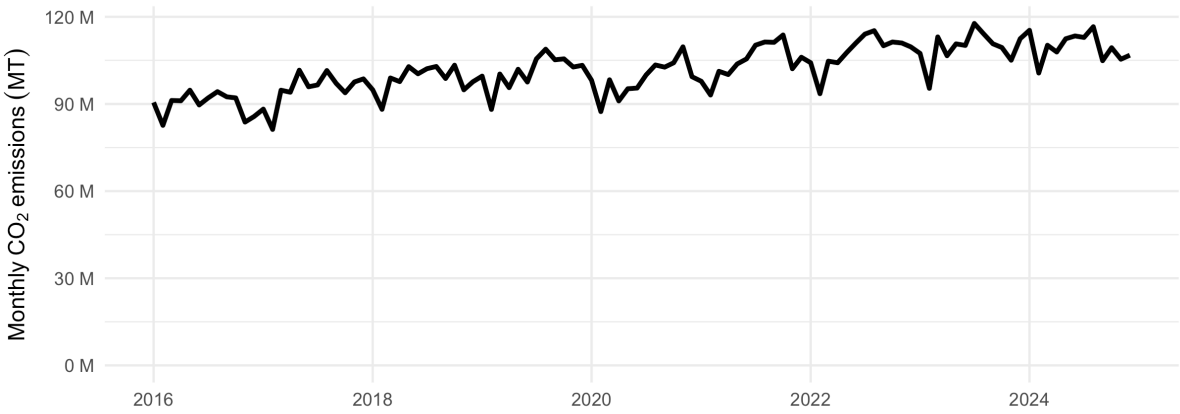

Figure 5: Time series of monthly global CO2 emissions, aggregated across AIS-broadcasting vessels and non-broadcasting vessels.

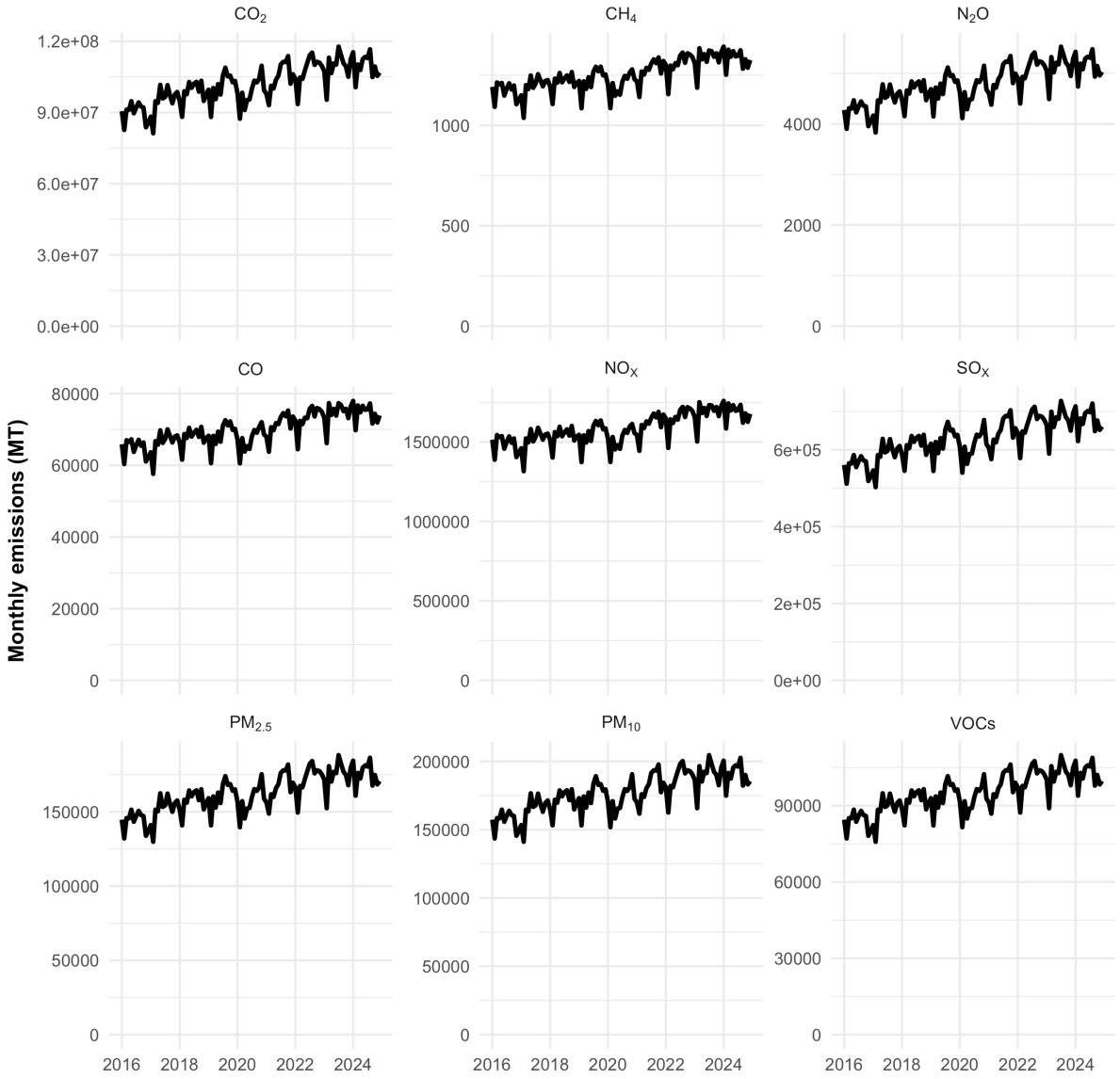

Figure 6: Time series of monthly global emissions for each pollutant, aggregated across AIS-broadcasting vessels and non-broadcasting vessels.

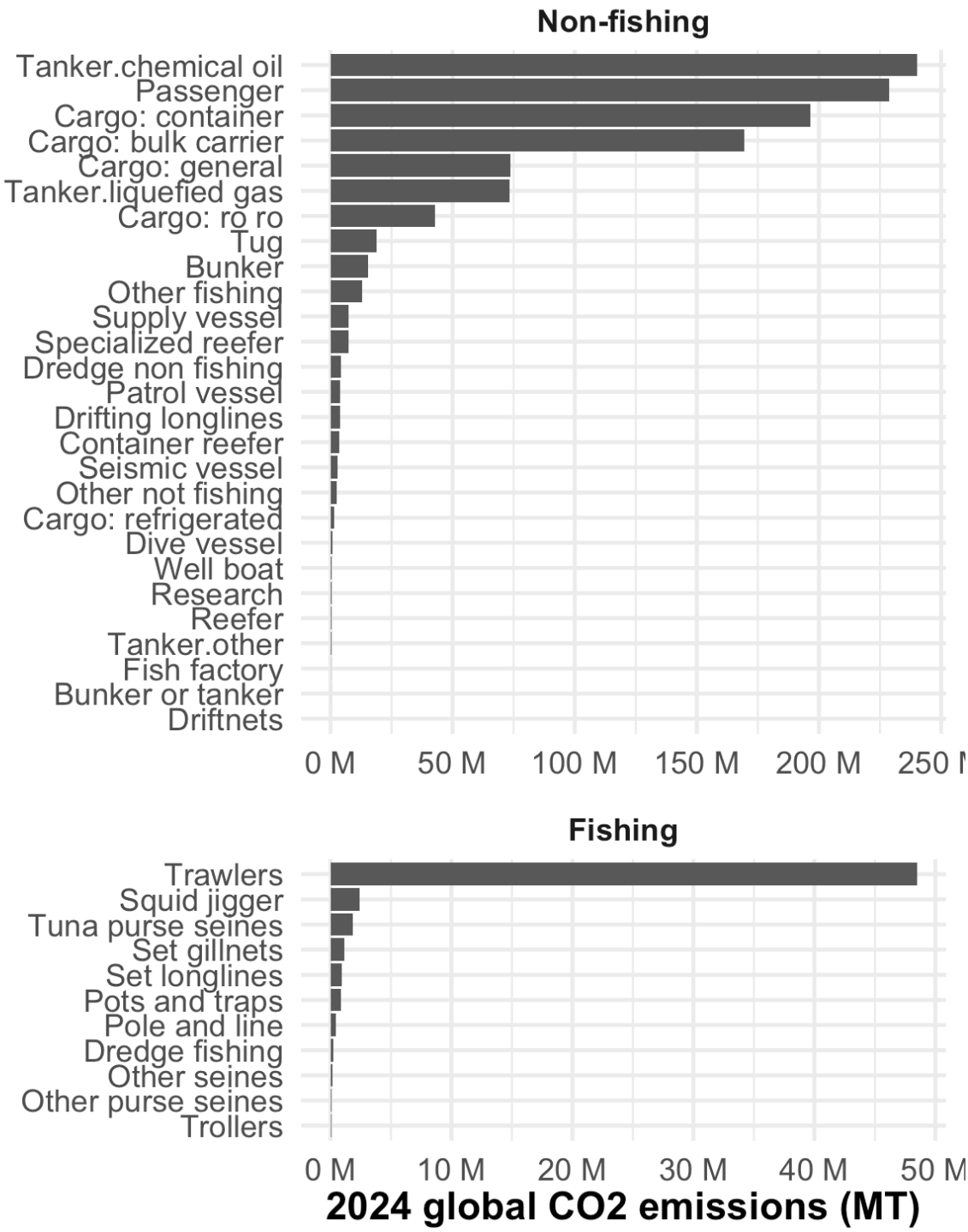

Figure 7: Total 2024 CO2 emissions by vessel type for AIS-broadcasting vessels (millions metric tonnes).